Exam Ace Champs
Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

PHYSICS

class 12

Covers
all chapters of NCERT

Faster Recall

Quick Revision

Class : 12th Physics
Chapter- 1 : Electric Charges and Fields

Electric Charges and Fields

Electric Charges

- Two kinds of charges + ve and – ve
- S.I unit is coulomb (C)
  1 C = charge flowing through
- a wire in 1 sec if current in it is 1A
  $1e = 1.602 \times 10^{-19}$C

Conservation of charge

It is not possible to create or destroy net charge of an isolated system.

Quantization of charge

All observable charges must be integral multiple of e. i.e.
Q = ne
where n = integers

Coulomb's Law

- Force between two charged particle

$$\vec{F} = \frac{k\, q_1 q_2 \vec{r}}{r^3} = \frac{k q_1 q_2 \hat{r}}{r^2}$$, force is attractive when q_1 and q_2 are of opposite sign, otherwise repulsive

$$|\vec{F}| = \frac{k\, q_1 q_2}{r^2}$$

$$k = \frac{1}{4\pi\varepsilon_0} \approx 9\times 10^9\,\mathrm{Nm^2C^{-2}}$$

- Force between multiple charges

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} + \ = \frac{q_1}{4\pi\varepsilon_0}\sum_{i=2}^{n}\frac{q_i}{r_{1i}^2}\hat{r}_{1i}$$

where $\hat{r}_{1i}$ is the unit vector along charge particle under consideration
$\hat{r}_{1i}$ = distance between I and i^{th} particle

"The electric force with which two charges attracts or repel one another are not affected by the presence of other charges". It is known as Superposition Principle.

Electric Flux

$\Delta\phi = \vec{E}.\Delta\vec{S}$, $\Delta\vec{S}$ = area vector
E = electric field
$\phi = \int \vec{E}.\vec{ds}$

Gauss's Theorem

$\oint \vec{E}.\vec{ds} = q_{in}/\varepsilon_0 = \phi$
where, q_{in} = net charge enclosed
ϕ = flux through a closed surface

Application of Gauss's Law

Electric field due to uniformly charged sphere at an
- Outside point
  $E = Q/4\pi\varepsilon_0 r^2$
  Internal point
  $E = Qr/4\pi\varepsilon_0 R^3$
- On the surface
  $E = Q/4\pi\varepsilon_0 R^2$

- Field due to linear charge distribution
  $E = \lambda/2\pi\,\varepsilon_0 r$,
  λ = linear charge density
- Field due to a plane sheet of charge
  $E = \sigma/2\varepsilon_0$,
  σ = charge per unit area
- Field due to a charged conducting plate
  $E = \sigma/\varepsilon_0$,
  σ = charge per unit area
- Field inside a conductor = 0
- Electric field due to a thin spherical shell at a point
  Outside the shell, $E = Q/4\pi\varepsilon_0 r^2$
  Inside the shell, E = 0
  On the surface $E = Q/4\pi\varepsilon_0 R^2$
- Electric field due to uniformly charged share at an
  Outside point
  $E = Q/4\pi\varepsilon_0 r^2$
  Internal point
  $E = Qr/4\pi\varepsilon_0 R^3$
  On the surface
  $E = Q/4\pi\varepsilon_0 R^2$

Electric field

Electric field lines

- Always normal to conducting surface - Lines originating from +ve charge - Terminating at - ve charge - Never intersect each other - Never from closed loop - Are imaginary lines.

Electric field due to point charge

$\vec{E} = \frac{\vec{F}}{q} \quad = k\,\frac{Q}{r^2}$ (N/C)

$\because$ unit = N/C

- Due to discrete distribution of charges
  $\vec{E} = \sum_{i}^{n}\vec{E}_i$
- Due to continuous distribution of charges
  $\vec{E} = k\int\frac{dQ}{r^3}\vec{r}$
  $|\vec{E}| = k\int\frac{dQ}{r^2}$

Electric Dipole, $\vec{p} = q\vec{d}$

- Electric field due to dipole at axial position
  $\vec{E} = \frac{1}{4\pi\,\varepsilon_0}\left(\frac{2P}{r^3}\right)\hat{r}$
- Electric field due to dipole at equatorial position
  $|E| = \frac{1}{4\pi\,\varepsilon_0}\left(\frac{P}{r^3}\right)$
- Torque on an electric dipole placed in an electric field $(\vec{E})$
  $\vec{\tau} = \vec{p}\times\vec{e} = p\,E\sin\theta$

Class : 12th Physics
Chapter- 2 : Electrostatic Potential and Capacitance

Electrostatic Potential and Capacitance

Electric Potential

Work done per unit test charge by an external agent in moving the test charge from reference point to the desired point S.I. unit J/C

V_0 = Work done/charge

Electric potential due to a point charge

$$V = \frac{1}{4\pi\varepsilon_0}\sum\frac{Q_i}{r_i}$$

Potential due to a system of charges

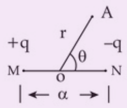
$$V_P = Q/4\pi\varepsilon_0 r$$

Potential due to a dipole

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

Where $p = qd$

$\theta = \angle AON$



At $\theta = 0$, $V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$ [axial position]

At, $\theta = 90°$, $V = 0$ [equitorial position]

Capacitance, $C = \frac{Q}{V}$

Capacitance of a parallel plate capacitor

$C = K\varepsilon_0 A/d$, K = dielectric constant

Capacitance when material slab inserted between them

$C = K\varepsilon_0 A/[Kd - x(K-1)]$

where x = thickness of the slab inserted

Parallel grouping of capacitors

$C = C_1 + C_2 + C_3 + ---- + C_n$

for two, $C = C_1 + C_2$

Capacitance of a spherical capacitor

$$C = 4\pi\varepsilon_0 / \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For isolated sphere

$C = 4\pi\varepsilon_0 R$

Energy stored in a capacitor

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$$

Series grouping of capacitors

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + --- + \frac{1}{C_n}$$

For two, $C = \frac{C_1 C_2}{C_1 + C_2}$

Dielectric

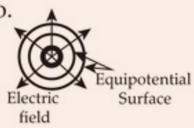
Conductors	Insulators
Such a material Which when placed in an electric field, the free electrons move in a direction opposite to the field.	Such a material in which electrons are tightly bound, & when exposed in an electric field, Electrons does not move *i.e.* having no free electrons

- Electric field inside a conductor is zero
- Electric field is always perpendicular to the charged surface
- In static state, there will be no additional charge in a conductor

- $F = \frac{q_1 q_2}{4\pi K\varepsilon_0 r^2}$, K = dielectric constant of medium
- A dielectric is an electrical insulator that can be polarized by an applied electric field.

Equipotential Surface

- Potential is same at all the points of the surface
- Component of electric field parallel to an equipotential surface is zero.



Electric Potential Energy

- It is negative of work done by the electric force as the configuration of the system changes.
- $U_{r_2} - U_{r_1} = -W = \frac{q_1 \cdot q_2}{4\pi\varepsilon_0}\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$
- If the separation between charges is 'r' then $U_{(r)} = \frac{q_1 q_2}{4\pi\varepsilon r}$
- Potential Difference,

$$V_B - V_A = \frac{U_B - U_A}{q}$$

$U_B - U_A$ = change in Potential Energy

q = test charge

Potential Energy of a dipole

$dU = pE\sin\theta d\theta$

If we choose P.E. of dipole to be zero when $\theta = 90°$ then

$$U_\theta - U_{90°} = \int_{90°}^{\theta} pE\sin\theta\, d\theta$$

$$U_\theta = -pE\cos\theta = -\vec{p}\times\vec{E}$$

If it is rotated through angle θ against the torque

Class : 12th Physics
Chapter- 3 : Current Electricity

Current Electricity

Current density
$\vec{J} = \dfrac{\Delta i}{\Delta s \cos\theta} = \sigma E$

Δs = area of cross-section

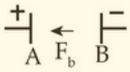
θ = angle between Area vector & current flow

σ = conductivity

EMF
it is a device which maintains a potential difference between its two terminals

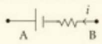
emf $= E = \dfrac{W}{q} = \dfrac{F_b}{d}$

It is equal to the potential difference between the terminals when the terminals are not connected externally.



Internal Resistance
$V_A - V_B = E - ir$

r = internal resistance of the cell



Grouping of Cell
Parallel grouping

$E_{eq} = \dfrac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$, $i = \dfrac{E_{eq}}{R + \dfrac{r_1 r_2}{r_1 + r_2}}$

Series grouping

$i = \dfrac{E_1 + E_2}{R + (r_1 + r_2)} = \dfrac{E_0}{R_1 + R_2}$

$E_0 = E_1 + E_2$

Electric Current
Rate of flow of charge through an area.

$i = \dfrac{Q}{t}$

$i = \int \vec{J}.\vec{ds}$

Temperature Dependence of Resistance
$R = R_0 [1 + \alpha (T - T_0)]$

$\rho = \int [1 + \alpha (T - T_0)]$

R = resistance at temperature T

R_0 = resistance at temperature T_0

α = coefficient of resistivity

Drift velocity
$V_d = \dfrac{i}{neA} = \dfrac{J}{ne}$

Mobility
$\mu = \dfrac{V_d}{E} = \dfrac{e\tau}{m}$

τ = average collision time

In 1827 Georg Simon Ohm gave his complete theory of electricity in his book "Die galvanische Kette, mathematisch bearbeitet

Ohm's Law
$J = \sigma E$

or

$V = IR$

Resistance
$R = \rho \dfrac{l}{A} \tau$

Resistivity
$\rho = \dfrac{1}{\sigma} \Omega\, m$

Conductivity
$\sigma = J/E$

Grouping of Resistance
Parallel grouping

$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + - - - + \dfrac{1}{R_n}$

For two resistors

$R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

Series grouping

$R = R_1 + R_2 + R_3 + - - - + R_n$

For two resistors

$R = R_1 + R_2$

Kirchhoff 's Law
Loop Law

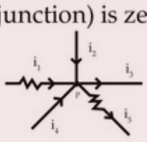
The algebraic sum of all the potential difference along a closed loop is zero.

Junction Law

$i_1 + i_2 + i_4 = i_3 + i_5$

algebraic sum of all the currents directed towards a point (junction) is zero.

$\Sigma I_p = 0$



Meter Bridge
Unknown resistance

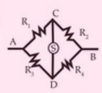
$S = \dfrac{R(100-l)}{l}$

l = balancing length

R = known resistance

Wheatstone Bridge
- is balanced

If $\dfrac{R_1}{R_2} = \dfrac{R_3}{R_4}$



- $V_c = V_D$

No current through G

Potentiometer
$P = \dfrac{V_A - V_B}{l} = \dfrac{\text{potential difference}}{\text{length of wire}}$

Comparison of emf, $\dfrac{E_1}{E_2} = \dfrac{l_1}{l_2}$

Internal Resistance (r) $= R\left(\dfrac{l_1}{l_2} - 1\right)$

$l_1\ l_2$ are lengths of null point

Class : 12th Physics
Chapter- 4 : Moving Charges and Magnetism

Moving Charges and Magnetism

Magnetic Force on moving charge

$\vec{F} = q\vec{v}\times\vec{B}$

$= qvB\sin\theta$

- For $\theta = 0$, $\vec{F} = 0$ along the magnetic field
- For $\theta = 90°$, i.e. if charge's velocity is perpendicular to field direction, Force is perpendicular to both field & velocity

$F = qvB = \frac{mv^2}{r}$

$r = \frac{mv}{qB}$ = radius of the circle in which charge rotates

Time period (T) $= \frac{2\pi m}{qB}$

ν(frequency) $= \frac{1}{T} = \frac{qB}{2\pi m}$

If $\theta \neq 0, 180°, 90°$

Then, $F = qvB\sin\theta$

And the charge particle will follow helix path whose

$r = \frac{m v_{\perp}}{q B}$ and pitch $= V_{\parallel} \times T = V_{\parallel} \times \frac{2\pi m}{qB}$

Magnetic Field $(\vec{B})$

- It is a region around a magnet or current carrying conductor or a moving charge in which its magnetic effect can be felt
- SI unit is Tesla(T) = weber/m^2
- 1 Gauss = 10^{-4} Tesla

Field due to a circular current

Field at the centre

$B = \frac{\mu_0 i}{2a}$ ⊗ ⊙

Field at an axial point

$B = \frac{\mu_0 i a^2}{2(a^2 + d^2)^{3/2}}$

Field at a point far away from the centre

$B = \frac{\mu_0 i a^2}{2 d^3}$

Biot- Savart Law

⊙ ⊗

$d\vec{B} = \frac{\mu_0}{4\pi}\frac{i d\vec{l}\times\vec{r}}{r^3}$

$dB = \frac{\mu_0}{4\pi}\frac{i dl\sin\theta}{r^2}$

where $\mu_0 = \frac{1}{\varepsilon_0 c^2}$

$= 4\pi \times 10^{-7}$ TmA^{-1}

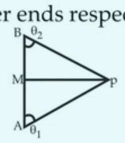
θ = angle between $d\vec{l}$ and $\vec{r}$

Direction of field will be perpendicular to plane containing current element and the point of observation.

Magnetic field due to straight wire current

$B = \frac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2)$

where θ_1 and θ_2 are the angle corresponding to the lower and upper ends respectively



Field due to a long straight wire current

i.e $\theta_1 = 0$

$\theta_2 = \pi$

$B = \frac{\mu_0 i}{2\pi d}$

Ampere's Law

$\oint \vec{B}.d\vec{l} = \mu_0 i$

where i = total current crossing the area bounded by closed curve.

Solenoid

- Magnetic filed at a point inside due to a long solenoid $B = \mu_0 ni$
- And at point on one end $B = \frac{\mu_0 ni}{2}$

where n = no. of turns per unit length along the length of solenoid.

Magnetic field due to Toroid

$B = \frac{\mu_0 Ni}{2\pi r}$, N = total no. of turns

i = current in toroid

Force on a current carrying conductor

$d\vec{F} = i\, d\vec{l} \times \vec{B}$, $F = ilB$

Force between parallel currents

$F = \frac{\mu_0 i_1 i_2}{2\pi d}$

Torque experienced by a loop in uniform magnetic field

$\vec{\tau} = MB\sin\theta\,\hat{n} = \vec{M}\times\vec{B}$

Sensitivity of moving coil galvanometer is nBA/k

Definition of Ampere

If two parallel wire kept 1 m apart, if $F = 2 \times 10^{-7}$, then current = 1A in each.

Oersted Law

In April, 1820, Hans Christian Oersted discovered that flow of current in a wire could deflect a nearby magnetic compass needle

Gal. to ammeter

$S = \frac{I_g}{I_T - I_g}G$

Gal. to voltmeter

$R = \frac{V}{I_g} - G$

IBRAIN

Class : 12th Physics
Chapter- 5 : Magnetism and Matter

Magnetism and Matter

Dipole moment

- **Revolving Electron**
  - $|\vec{\mu}| = iA = \frac{ev}{2\pi r} \times \pi r^2$
  - $= evr/2$
  - $\vec{\mu} = \frac{-e}{2m_e}\vec{L}$
  - $\mu = md$
  - m = magnetic charge or pole strength
  - d = separation between poles
  - r = radius of circular orbit
  - e = charge on electron
  - v = speed of electron;
  - L = angular momentum
  - m_e = mass of electron
- **Circular Loop**
  - Current loop of area $A = \pi r^2$ carrying current i may be replaced by a magnetic dipole of dipole moment $\mu = md = iA$
- **Solenoid**
  - $\mu = NIA$,
  - N = no. of turns
  - I = current
  - A = area

Earth's Magnetic field

Earth is a natural source of magnetic field having geometric north and geometric south pole

- **Declination**
  - Angle b/w magnetic meridian and geographical meridian.
- **Inclination or dip (δ)**
  - Angle made by earth's magnetic field with horizontal in the magnetic direction
  - $\delta_{equator} = 0, \delta_{pole} = 90^\circ$
- **Horizontal component**
  - $B_H = B\cos\theta$
  - B_H (at pole) $= 0$
  - B_H(at equator) is maximum

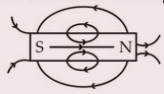
Magnetic Substance

- **Para-magnetism**
  - Tendency to increase the magnetic field due to magnetization
  - e.g. Al, Mn I, X_m, $\mu_r > 1$
- **Diamagnetism**
  - Tendency to magnetise in a direction opposite to direction of magnetic field
  - e.g. Bi, Cu, Hg, Ni. μ_r, I, χ_m
- **Ferromagnetism**
  - Tendency of strongly magnetization in the direction of magnetic filed.
  - e.g. Fe, Co, Ni, μ_r, I, X_m >>1

In 1833, Carl Friedrich Gauss and Wilhelm Weber Made the first Electromagnetic Telegraph

Magnetic field lines

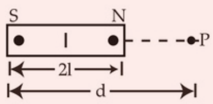
Imaginary lines which continuously represent the direction of magnetic field.

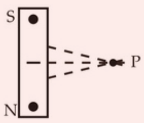


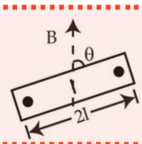
Electromagnets

Magnet in which the magnetic field is produced by an electric current. Magnetic field disappears when current is turned off.

Bar Magnet

- **Magnetic field Intensity**
  - End–on–position
    $$= \frac{\mu_0}{4\pi}\frac{2M}{d^3}$$
    $M = 2ml$
    $d >>> l$
    

  - Broadside –on Position
    $$B = \frac{\mu_0}{4\pi}\frac{M}{(d^2+l^2)^{3/2}}$$
    For $d >>> l$
    $B = \mu_0 M/4\pi d^3$
    

- **potential Energy of a Field- Magnet System**
  - $U_\theta = U_\theta - U_{90^\circ} = -MB\cos\theta = -\vec{M}.\vec{B}$
  - we take P.E. at $\theta = 90^\circ$ to be zero
- **Torque**
  - $\vec{\tau} = \vec{M} \times \vec{B} = MB\sin\theta$
  - $M = 2ml$
  - B = Magnetic field
  

Magnetic Terms

- **Magnetic Intensity**
  - $H = \frac{B_0}{\mu}$
- **Magnetic permeability**
  - $\mu = \mu_0(1+\chi_m)$
  - $\mu = \mu_0(1+\chi_m)$
- **Time period oscillating bar magnet**
  - $(T) = 2\pi\sqrt{\frac{I}{MB}}$
- **Magnetic susceptibility**
  - $\chi = \frac{I}{H}$
- **Intensity of magnetization**
  - $I = \frac{M}{V}$

Class : 12th Physics
Chapter- 6 : Electromagnetic Induction

Electromagnetic Induction

Self Inductance of long Solenoid

$L = \mu_0 n^2 \pi r^2 l$

n = no. of turns per unit length

ϕ = flux = $(\mu_0 n i)\, \pi r^2$

r = radius of each loop of solenoid

- Growth of current in LR Circuit

  $i = \frac{E}{R}(1 - e^{tR/L}) = i_0 (1 - e^{-t/\tau})$

- Decay of current

  $i = i_0 e^{-t/\tau}$

- Energy stored in an Inductor

  $U = \frac{1}{2} L i^2$

Faraday's Law of Electromagnetic Induction

Whenever flux of magnetic field through the area bounded by a closed conducting loop changes, an emf is produced in the loop

The emf is given by $E = -d\phi/dt$

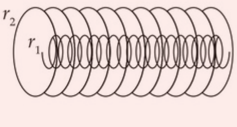
where $\phi = \int \vec{B}.\vec{ds}$

is the flux of the magnetic field through the area.

Mutual Inductance

$\phi = Mi$

$\frac{d\phi}{dt} = -M\frac{di}{dt}$



$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$

$M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l$

Emf induced in an AC generator, $E = NBA\omega \sin \omega t$

Lenz's Law

The direction of the induced current is such that it opposes the changes that has induced it.

Self Induction

If a solenoid of N turns, the flux through each each turn $= \phi = \int \vec{B}.\vec{ds}$

emf induced between the ends of coil $= E = -N\frac{d}{dt}\int \vec{B}.\vec{ds}$

Contribution

In 1831, Michael Faraday discovered electromagnetic induction and James Clerk Maxwell mathematically described it.

Induced current

$I = \frac{E}{R} = -\frac{1}{R}\frac{d\phi}{dt}$

Eddy Current

It is induced when magnetic flux linked with the conductor changes

Induced EMF

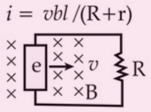
$E = \frac{d\phi}{dt}$

Motional EMF

$E = \left|\frac{d\phi}{dt}\right| = \left|Bl\frac{dx}{dt}\right|$

$= Blv$

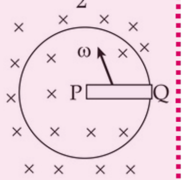
$i = vbl/(R+r)$



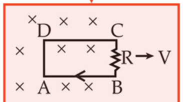
r = resistance of rod moving with velocity v in uniform magnetic field $\vec{B}$

EMF induced in a rotating conductor

$E = \frac{1}{2}B\omega l^2$



Rectangular loop



emf induced

$E = vBl$

$i = \frac{vBl}{R}$

Magnetic force on the loop

$F = B^2 l^2 v/R$

= force required to move the loop with constant velocity (v)

Thermal power developed in the loop is

$P = \frac{v^2 B^2 l^2}{R}$

Class : 12th Physics
Chapter- 7 : Alternating Current

Alternating Current

AC Circuits

Inductive (L) circuit
$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$
Phase difference b/w V & I, $\phi = \pi/2$
Power factor $\cos(\phi) = 0$
Voltage leads current by $\pi/2$
$P = 0$

Capacitive (C) circuit
$i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$
$\phi = \pi/2$ or $-\pi/2$
$\cos\phi = 0$
Power(P) $= 0$
Current leads the voltage by $\pi/2$

Resistive (R) circuit
$i = i_0 \sin(\omega t)$
$\phi = 0°$
$\cos\phi = 1$
$P = V_0 I_0 / 2$
Current & voltage both are in same phase

R- C circuit
$i = i_0 \sin(\omega t + \phi)$
$Z = \sqrt{R^2 + X_c^2}$
$X_c = 1/\omega C$
$\tan\phi = 1/\omega CR$
$i_0 = E_0/Z$
- Power factor $\cos\phi = \frac{R^2}{\sqrt{R^2 + X_c^2}}$
- current leading.

L- C circuit
$i = i_0 \sin\left(\omega t \pm \frac{\pi}{2}\right)$
$X = X_L - X_C$
$\phi = \pi/2$
- Power factor $= 0$, Either voltage or current leading

L- R circuit
$i = i_0 \sin(\omega t + \phi)$
$Z = \sqrt{R^2 + X_L^2}$
$X_L = \omega L$
$\tan\phi = \omega L/R$
$i_0 = E_0/Z$
- power factor $= R/Z$
- voltage is leading

L- C- R circuit
$i = i_0 \sin(\omega t \pm \phi)$
$Z = \sqrt{R^2 + (X_c - X_L)^2}$
$X = X_c - X_L$
$= \left(\frac{1}{\omega c} - \omega L\right)$
$\tan\phi = \left(\frac{1}{\omega c} - \omega L\right)/R$, $i_0 = \frac{E_0}{Z}$

Impedance
The ratio of emf applied and the current produced in an AC circuit
$Z = E_0/i_0$ Ohm

Reactance
The hindrance offered by inductance or capacitance or both to the flow of AC in AC denoted by (X) Ohm

Contribution
In 1883 Nikola Tesla constructed the first induction motor at the age of 27 years only.

Power in AC Circuit
$P = E_{rms}\, i_{rms} \cos\phi = \frac{E_0}{\sqrt{2}} \times \frac{i_0}{\sqrt{2}} \cos\phi = \frac{1}{2} E_0 i_0 \cos\phi$

AC Current
Direction of current changes alternatively
$i = i_0 \sin\omega t$
$v = v_0 \sin\omega t$

RMS value or root mean square value
$i_{rms} = i_0/\sqrt{2}$
$v_{rms} = v_0/\sqrt{2}$

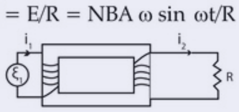
Mean value or Average value
$i_{mean} = 2i_0/\pi$
$v_{rms} = 2v_0/\pi$

Peak value or Max^m value
$i_0 = \sqrt{2}\, i_{rms}$
$v_0 = \sqrt{2}\, v_{rms}$

Transformer
It is used to obtain a high AC voltage from a Low- voltage AC source or vice-versa

Efficiency of a transformer
- η = output power/Input power
- 99% efficiency can easily achieved

AC Generator
A device generating alternating current having elements magnet, Armature and graphite Brushes emf induced through coil
$E = NBA\,\omega \sin\omega t$
$i = E/R = NBA\,\omega \sin \omega t/R$



$E_2 = -N_2E_1/N_1$
$i_2 = -N_1E_1/N_2$
If $N_2 > N_1$ it is called step-up Transfomer
If $N_1 > N_2$, it is called step-down Transfomer

Class : 12th Physics
Chapter- 8 : Electromagnetic Waves

Electromagnetic Waves

Maxwell's Equation

Gauss's Law (electricity)

$$\oint \vec{E}.\vec{ds} = \frac{q_{in}}{\varepsilon_0}$$

Gauss's Law (magnetism)

$$\oint \vec{B}.\vec{ds} = 0$$

Faraday's Law

$$\oint \vec{E}.\vec{dl} = \frac{-d\phi_B}{dt}$$

Ampere's Law

$$\oint \vec{B}.\vec{dl} = \mu_0 i + \mu_0\varepsilon_0 \frac{-d\phi_E}{dt}$$

Displacement Current

If there exists an electric current as well as a changing electric field, results magnetic field & cause displacement current

$$i_d = \varepsilon_0 \frac{-d\phi_E}{dt}$$

Electromagnetic spectrum

- Radio wave $\lambda > 10^8$nm
  Use : radio, TV signal
- Micro wave $10^8 > \lambda > 10^5$nm
  Use : micro wave oven, radar
- Infrared $10^5 > \lambda > 700$ nm
  Use : night vision
- Visible light $700 > \lambda > 400$ nm
  Use : to observe world
- UV rays $400 > \lambda > 10$nm
  Use : destroying bacteria
- X-rays $10 > \lambda > 0.01$ nm
  Use : detect bone break
- γ rays $0.01\text{nm} > \lambda$
  Use : to treat cancer

Energy density

- $U = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}$
  = energy per unit volume
- U_{av} over a long time

$$= \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\frac{B_0}{\mu_0}$$

$$= \frac{1}{2}\varepsilon_0 E_0^2 \ [\text{as } c = B/E_0 \ \& \ \mu_0\varepsilon_0 = \frac{1}{c^2}]$$

$$= \frac{1}{2}\frac{B_0^2}{\mu_0}$$

Conduction current

The current due to flow of charge is often called

$$i_c = \frac{d\theta}{dt}$$

Contribution

In 1886 Henrich Hertz became the first person to transmit and received controlled radio waves

Combination of mutually perpendicular electric & magnetic fields is referred to as an electromagnetic wave

Characteristics of Electromagnetic waves

1. Do not need any material medium for propagation
2. Travels with speed (v) $= \frac{1}{\sqrt{\mu_0\varepsilon_0}}$
3. Produced by accelerated charge
4. Transverse in nature
5. Oscillating electric & magnetic fields are in phase and their ratio is constant $(C) = E_0/B_0$

Class : 12th Physics
Chapter- 9 : Ray Optics and Optical Instruments

Ray Optics and Optical Instruments

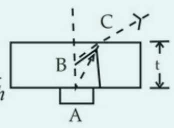
Total Internal Refection

When ray passes from optically denser to rarer medium. If incident angle (i) further increased than $(\theta)_C$ critical angle entire light is then reflected back to the denser medium again is called T.I.R. It is used in optical fibre.

Refraction of light

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$$

$$\mu = \frac{real\ depth}{apparent\ depth}$$



$$\Delta t = \left(1 - \frac{1}{\mu}\right)t = \text{image shift}$$

Spherical Mirror

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f}$$

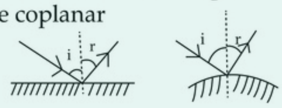
Lateral Magnification $= \frac{h_2}{h_1} = -\frac{v}{u}$

Sign Conversion

- Pole is taken as origin
- Principle axis as the X-axis
- All distance measured from origin (or pole).
- All distance measured in the direction of incident ray is taken + ve.
- All distance measured in the direction opposite to the incident ray is taken - ve.

Reflection of light

- $\angle i = \angle r$
- Incident ray reflected ray and normal to the reflecting surface are coplanar

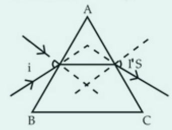


Prism

Angle of deviation $\delta = i - i' - A$

$\delta_{minimum} = 2i - A\,[i = i'] \rightarrow$

$\delta_{minimum} = (\mu - 1)A$, if A is small



Lens Maker's Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{\mu_2 - \mu_1}{\mu_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad [If\ \mu_2 = \mu, \mu_1 = 1\ (air)]$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \text{(lens formula)}$$

Lateral Magnification $= \frac{h_2}{h_1} = \frac{v}{u}$

Critical Angel

- Incident angle (θ_c) for which angle of refraction is 90°

i.e. $\sin\theta_c = 1/\mu$

$$\theta_c = \sin^{-1}\left(\frac{1}{\mu}\right)$$

When ray passes from optically denser to rarer medium.

Refraction on spherical surface

$\mu_1 \ (\mu_2$

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Lateral Magnification

$$m = \frac{h_2}{h_1} = \frac{\mu_1 v}{\mu_2 u} = \frac{R - v}{R - u}$$

Scattering of light

Light scattered i.e. redirected in different paths when interacts with particle matters e.g. sunset, sunrise, colours, blue colour of sky

Power of a lens

$$P = \frac{1}{f}$$

[For combination of lens]

$$P = \frac{1}{f_1} + \frac{1}{f_2}$$

Optical Instruments

Optical Instruments

$M = 1 + \frac{D}{f}$ [image at near point]

$M = D/f$ [image at infinite]

Optical Instruments

$M = \frac{v}{u}\left[\frac{D}{f_e}\right]$ [normal adjustment]

$M = \frac{v}{u}\left(1 + \frac{D}{f_e}\right) = -\frac{1}{f_o}\left(1 + \frac{D}{f_e}\right)$

For final image at least distance

Optical Instruments

$M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$ [image at near point]

$M = -\frac{fo}{fe}$ [image at infinite]

Class : 12th Physics
Chapter- 10 : Wave Optics

Wave Optics

Huygen's Principle
- Each point on the primary wavefront is the source of a secondary wavelets
- **Wave front**
  - Locus of all particles vibrating in same phase
  - For spherical wavefront
    - $I \propto \frac{1}{r^2}$
    - $A \propto \frac{1}{r}$

Resolving Power
- Resolving power of microscope $= 2n\sin\theta/\lambda$
- Resolving power of telescope $= d/1.22\lambda$

Polarisation
- Restricting the vibration of light in a particular direction perpendicular to the direction of propagation of wave
- **Brewster's Law**
  - $\mu = \tan\theta_p$
  - θ_p = angle of polarisation
  - Polariod used in lab to analyse plane polarised light
  - used to eliminate the head light glare in motor cars.

Young's Double Slit Experiment
- **Fringe- width**
  - $\beta = \frac{D}{d}\lambda$ is a distance between two consecutive bright or dark fringe
- **Based on interference of light**
  - Distance between nth bright fringe and central fringe
    $x_n = \frac{n\lambda D}{d}$
    D= distance between source and screen
  - Distance between nth dark fringe and central fringe
    $x_n = \frac{(2n-1)\lambda D}{2d}$
    d = distance between slits
- **Path difference**
  - For bright fringe $\Delta\Phi = n\lambda$
  - For dark fringe $\Delta\Phi = \left(n + \frac{1}{2}\right)\lambda$

Diffraction of light by single slit
- $b\sin\theta = \eta\lambda$ (dark fringe)
- Linear width of central maxima
- Width of central maxima $= \frac{2\lambda}{b}$
- Angular width $= \frac{2D\lambda}{b}$
- $b\sin\theta = (2n+1)\frac{\lambda}{2}$ (for maxima bright fringe)

Coherent source
- Two sources of light is said to be coherent if the initial phase difference between the waves emitted by the sources remains constant in time otherwise they are called Incoherent source of light

Interference of light
- Two wave superpose to form a resultant wave of greater or lower or same amplitude
- $I = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$
  $= \beta(A_1 + A_2)^2$
- For constructive Interference $A = A_1 + A_2$
- For destructive interference $A = (A_1 - A_2)$

Dual nature of radiation and matter

Dual Nature of Radia

- Light has both wave character as well as particle
- Interference explained by wave nature
- When light of sufficiently low wavelength, it behaves as a particle
- Light particle having definite energy and definite linear momentum called "photon" or Energy packets.

Energy of each photon = $h\nu = hc/\lambda$

Momentum of each photon = $h/\lambda = E/c$

Contribution

Einstein, after an average academic career put forward quantum theory of light in 1905 while working as a grade III technical officer in a patent office

Matter Waves

All matter can exhibit wave–like behaviour e.g. beam of electrons can be diffracted like water wave

Davisson- Germer Experiment

A beam of electrons emitted by electron gun is made to fall on nickel crystal cut along cubical axis at a particular angle .Scattered beam of electrons is received by detector.

Results : λ = de Broglie's wavelength

$= h/p$

$= 1227/\sqrt{v}$ nm

$= 1227/\sqrt{54}$ nm

$= 0.167$ nm $= 1.67$ A°

This experiment verifies the wave nature of electrons & relation with de-Broglie wavelength.

De - Broglie relation

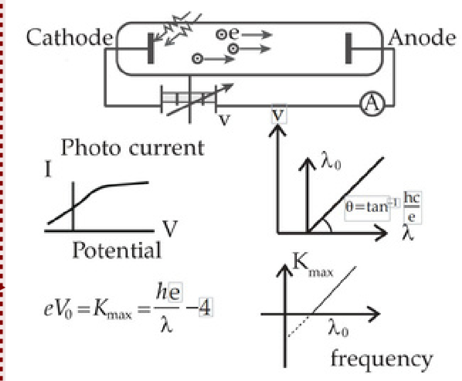
$\lambda = h/p$

λ = wavelength associated with particle or de-Broglie wavelength

p = momentum

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2\,m\,K_{max}}}$$

Hertz and Lenard's Observation



$eV_0 = K_{max} = \frac{hc}{\lambda} - \phi$

Photoelectric effects

- When light of sufficiently small wavelength is incident on metal surface, electrons are ejected from the metal, the phenomenon is called photoelectric effect.
- Ejected electrons are called photoelectrons
- Minimum energy equal to work function (ϕ) must be given to an electron so as to bring it out of the metal

Einstein 's photoelectric equation

- If $\lambda = \lambda_0 = hc/\phi$
  $K_{max} = 0$, *i.e.*
  Electron may just come out.
- If $\lambda > \lambda_0$
  i.e. $E < \phi$
  no electron will come out
- If $\lambda \le \lambda_0$.
  Photoelectric effect takes place this λ_0
- λ_0 = depends on metal used

- $K_{max} = E - \phi = eV_0$
  $= \frac{hc}{\lambda} - \phi$, V_0 = stopping potential
  K_{max} = maximum kinetic energy of elected electrons

Hore, $\lambda_0 = hc/\phi$

λ_0 = Threshold Wavelength

$\lambda_0 = c/\lambda_0 = \phi/h$

λ_0 = Threshold frequency

$K_{max} = \lambda(\nu - \nu_0)$

Class : 12th Physics
Chapter : 12 Atoms

Atoms

Contribution

In 1898 J.J Thomson proposed the first model of atom also known as plum-pudding model

Energy Level

Energy of electron in each stationary orbits is

Given by $E_n = -13.6\,\frac{Z^2}{n^2}\,eV$

Where $n = 1,2,3 \ldots$

Z = atomic number of atom

- These stationary energy orbits are also called energy levels.
- When electron jumps from higher energy level to lower energy level it releases energy.

$$\Delta E = E_f - E_i = 13.6\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]Z^2 eV$$

Postulates

- Atoms have a central, massive, positively charged core called nucleus around which electron revolves
- Size of nucleus

≈ 1 fermi $= 10^{-15}$m

Rutherford's Modal
Draw backs

- Doesn't explain the stability of atom
- Doesn't explain the atomic s ectrum

a - particle scattering Experiment

- α- particle bombarded on thin gold foil
- Most of α– particle passed undeviated or with a small angle
- 1 out of 8000 α particles were deflected by scattering angle

Impact parameter

$$b = \frac{Ze^2 \cot\theta/2}{4\pi\varepsilon_0\left(\frac{1}{2}mv^2\right)}$$

Hydrogen Spectrum

Hydrogen gas heated in a sealed tube emits radiation which passes through prism

- Component of different wavelength appears
- Wavelength in each series given by

$$\bar{v} = \frac{1}{\lambda} = R\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]$$

$n_f > n_i$

- Lyman series [U.V. region]

  $n_i = 1$, $n = 2,3,4 \ldots$

  $\lambda_{min} = 912$ A°, $\lambda_{max} = 1216$ A°
- Balmer series [visible region]

  $n_i = 2$ $n_f = 3, 4, 5 \ldots$

  $\lambda_{min} = 3648$ A°, $\lambda_{max} = 6563$ A°
- Paschan series [I–R region]

  $n_i = 3$ $n_f = 4, 5, 6 \ldots$

  $\lambda_{min} = 8208$ A°, $\lambda_{max} = 18761$ A°
- Brackett series [I R region]

  $n_i = 4$ $n_f = 5, 6, 7 \ldots$

  $\lambda_{min} = 14592$ A° $\lambda_{max} = 40533$ A°
- P –fund [I R region]

  $n_i = 5$ $n_f = 6, 7, 8 \ldots$

  $\lambda_{min} = 23850$ A° $\lambda_{max} = 74618$ A°

Bohr Modal

Postulates

- Electron revolves around the nucleus in stationary orbits
- Angular momentum of electron

  $$mvr_n = n \times \frac{h}{2\pi}$$

  n = Integers. It is also known as principle Quantum Number
- It explains spectrum of hydrogen or hydrogen like [He^+, Li^{++}] atom

Limitations

- Fails to explain spectrum of complex atoms/ions of multi electron system
- Dosn't explain Zeeman's and Stark's effect

Radius of nth Bohr's orbit

$$R_n = \frac{\varepsilon_0 h^2}{\pi m e^2}\left(\frac{n^2}{Z}\right)$$

$$= 0.53\,\frac{n^2}{Z}\,A°$$

Velocity of electron in nth Bohr's orbit

$$v_n = \frac{e^2}{2\varepsilon_0 h}\left(\frac{Z}{n}\right)$$

$$= 2.2 \times 10^6\,\frac{Z}{n}\ m/s$$

Potential Energy

$$U_n = -\frac{KZe^2}{r_n}$$

$$= \frac{-me^4}{4\varepsilon_0^2 h^2}\left(\frac{Z^2}{n^2}\right)$$

$$= \frac{-27.2\,Z^2}{n^2}\,eV$$

Kinetic Energy

$$E_k = \frac{KZe^2}{2r_n}$$

$$= \frac{me^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2}$$

$$= \frac{13.6^2}{n^2}Z^2 eV$$

Total Energy

$$E = U_n + E_k$$

$$= -\frac{KZe^2}{2r_n}$$

$$= \frac{-13.6\,Z^2}{n^2}\,eV$$

$$= \frac{U_n}{2} = -E_K$$

Class : 12th Physics
Chapter : 13 Nuclei

Nuclei

Composition
- Nuclei consists of protons and neutrons.
- Total No. of protons & neutrons in the nucleus is called nucleons
- ${}_ZX^A$, $A = Z + N$ = Mass Number
  A = Atomic Number
  N = No. of neutrons

Contribution
In 1896, Henri Becquerel discovers Radioactivity

Mass Energy Reaction
Einstein shows that mass is an another form of energy and gives a relation $E = mc^2$

Mass Defect
$\Delta M = [Z m_p + (A - Z)m_n] - M$
m_p = mass of proton, m_n = mass of neutron
M = mass of nucleus

Binding Energy
- $B.E. = \Delta Mc^2 = [Z m_p + (A - Z)m_n - M] \times c^2$
- B.E. per nucleon = B.E/A

Nuclear Fission
- Splitting of a heavy nucleus into two or more lighter nuclei.
  ${}_{92}U^{235} + {}_0n^1 \to {}_{56}Ba^{141} + {}_{36}Kr^{92} + 3{}_0n^1 + 200$ mev

Nuclear Fusion
- Combining two lighter nuclei to form a heavy nucleus
  ${}_1^1H + {}_1^1H \to {}_1^2H + e^+ + \nu + 0.42$ MeV (energy)

Types of Nuclei
- **Isotone**: Atoms whose nuclei having same number of neutron (N) e.g. ${}_{11}Na^{23}$, ${}_{12}Mg^{24}$
- **Isotopes**: Same atomic No.(Z), Different mass No.(A), e.g. ${}_6C^{12}$, ${}_6C^{14}$
- **Isobars**: Same mass No.(A), Different atomic No.(Z), e.g. ${}_6C^{14}$, ${}_7N^{14}$

Size of Nuclei
Nuclear Radius
$R = R_0A^{1/3}$
$R_0 = 1.2 \times 10^{-15}$ m
A = Mass No.

Nuclear Volume
$V = \frac{4}{3}\pi R^3$
$= \frac{4}{3}\pi R_0^3 A$
$V \propto A$ (mass no.)

Nuclear Density
$\rho = \frac{\text{mass of nucleus}}{\text{volume of nucleus}}$
$= \frac{A \times 1.67 \times 10^{-27}}{\frac{4}{3}\pi \times (1.2 \times 10^{-15})^3 \times A}$
$\approx 2.9 \times 10^{17}$ kgm^{-3} (independent of A)

Properties of α β & γ
α - Particle
- is a heavy doubly ionized helium ion
- least penetrating power
- highest ionizing power
- decay of 1 α particle cause decrease in atomic no. by 2 and mass no. by 4.

β - Particle
- is an electron moderate penetrating power (100 times α)
- less ionizing power than α –particle
- one β – particle decay cause increase in atomic no. by 1 and mass no. remains uncharged.

γ Rays
- A wave having high frequency having no mass,
- least ionizing power but highest penetrating power
- No charge in atomic No. and Mass No. on γ decay.

Radioactivity
Disintegration of heavy elements into comparatively lighter elements by emission of α β and γ radiations

Law of radioactivity
$-\frac{dN}{dt} = \lambda N$
$N = N_0 e^{-\lambda t}$

Half life
$t_{\frac{1}{2}} = \frac{0.693}{\lambda}$
$N = N_0\left(\frac{1}{2}\right)^{t/t_{\frac{1}{2}}}$

Mean Life
$\tau = \frac{1}{\lambda}$
$= t_{\frac{1}{2}} / 0.693$

Class : 12th Physics
Chapter- 14 : Semiconductor Electronics Materials, Devices & Simple Circuits

Semiconductor Electronics Materials, Devices and Simple Circuits

Classification of metals, semiconductor & Insulator on the basis of bond theory

Semiconductors

→ Eg = 0.72 eV for Ge
≈ 1.1ev for Si
$0 < E_g <$ Insulator

→ very low (e)

• For– n–type
$= 7 \times 10^{15} m^{-3}$
for *Si* doped with P
C.B
E_g
V.B

• For– p–type
$= 1 \times 10^{9} m^{-3}$
for Si doped with Al

Conductors

→ Eg = 0 or least
→ Density of charge carriers is very high (For electrons e)
$= 9 \times 10^{23}\ m^{-3}$ for For Hole (h)
$= 0\ m^{-3}$ (in case of copper)
C.B. V.B. $E_g = 0$

Insulator

→ Eg = very high ≈ 7eV for diamond
→ negligible ≈ 0
C.B
$E_g >>$
V.B

Energy Band

A range of energies associated with the quantum states of electrons in a crystalline solid

- **Valence Band**: The energy band which are completely filled with electron at OK.
- **Conduction Band**: The bands with higher energy above the valence band (C.B, E_g = Band gap, V.B)
- **Band Gap or Forbidden Gap**: The difference between the highest energy in a valence band and lowest energy in the next higher band.

Huygen's Principle

- **P.N junction Diode** → Device made by a close contact of N-type and P- type semiconductor
- **Forward Biased** → - +ve terminal to P-side; - - ve to N- side; - diffusion current increase
- **Reverse Biased** → - +ve terminal to P- side; - +ve terminal to N- side; - diffusion current decreases; - depletion layer increases
- **Light Emitting Diode/LED** → Used in T.V. or electronic gadgets
- **Photo Diode** → is a P.N. junction whose function is controlled by the light allowed to fall on it
- **Zener Diode** → Used as a voltage regulator

Contribution

In 1972, Walter Heitler and Fritz London discover bands

Intrinsic Semiconductor

- Pure semiconductor
- Charge carrier concentration
$n_i = n_e = n_h$
$I = n_e + n_h$
e.g. Si (Silicon)

Extrinsic Semiconductor

Impure or doped semiconductor

N-T type semiconductor

- Si or Ge doped with trivalent (B, Al) element
- Electrons are minority carriers
- Holes are majority carriers

P-T type semiconductor

- Si or Ge doped with pentavalent (P,As, Sb) element
- Electrons are majority carriers
- Holes are minority carriers

Transistor

is a three terminal device

p- n- p transistor

- In normal operation emitter, base junction is always forward biased

E base C
p n p
p-n-p transistor

n- p- n transistor

$I_E = I_B + I_C$
$\alpha = I_C/I_E$
$\beta = I_C/I_B$
$\beta = \dfrac{\alpha}{1-\alpha}$

E base C
n p n
n-p-n transistor

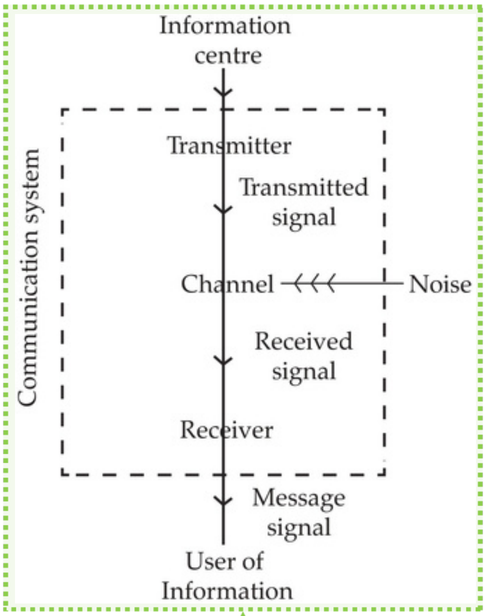
Logic Gates

1-Or Gate: A, B → Y=A+B
2-AND Gate: A, B → Y=A.B
3-NOT Gate: A → $Y=\overline{A}$
4-NOR Gate: A, B → $Y=\overline{A+B}$
5-NAND Gate: A, B → $Y=\overline{A.B}$

Class : 12th Physics
Chapter- 15 : Communication Systems

Communication Systems

Block Diagram



Band Width

It refers to the data carrying capacity of a channel or medium

Signal	Band width
Mobile phone	896 – 901 M Hz 840 – 935 M Hz
Satellite	5.9 – 6.4 G Hz
Communication	3.7 – 4.2 G Hz
AM	540 – 1600 kHz
FM	88 – 108 M Hz
TV (V HF)	54 – 72 M Hz
(V HF)	76 – 216 M Hz 420 – 890 M Hz

Propagation of Electromagnetic waves in Atmosphere

Transmitter antenna radiates E.M. waves, these waves travel through space & reach receiving antenna at other end.

Space satellite

Transmitter Antenna

Receiver Antenna

Contribution

Sir Tim Burners- Lee invented the world-wide web (www) in 1989

Sky waves

To transmit the information signal of low frequency, it is super imposed on high frequency wave as a carrier wave called modulation

Demodulation

The process of retrieval of information from the carrier wave at the receiver is termed as demodulation. It is reverse of modulation

Sky waves

- Ionosphere plays major role in sky wave propagation.
- In ionosphere ionization occurs due to the absorption of UV rays coming from the sun by the air molecule.
- In this layer bending of E.M. wave occurs so that they are diverted towards the earth which is helpful in sky wave propagation.
- Radio waves (1710 K HZ to 40 MHZ) are propagated in sky wave propagation.

Band width of Transmission Medium

Signal	Band width
Mobile phone	896 – 901 M Hz 840 – 935 M Hz
Satellite	5.9 – 6.4 G Hz
Communication	3.7 – 4.2 G Hz
AM	540 – 1600 kHz
FM	88 – 108 M Hz
TV (V HF)	54 – 72 M Hz

Space Wave

- Space waves travel in straight line from transmitter antenna to receiver antenna.
- It is used for line of sight communication as TV broadcast or satellite communication.
- The maximum line of sight (LOS) distance $d_m = \sqrt{2Rh_T} + \sqrt{2Rh_r}$
  h_T = height of transmitter antenna
  h_r = height of receiver antenna.